

Review retrospective:
*the general- h susceptibility, and a dead-open flag that was
wrong the other way*

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

2026-06-05

Abstract

A soundness review of `Laplace/OneD/AnharmonicSusceptibilityGeneralH.lean`: the susceptibility $\partial_h \langle u \rangle_h|_{h_0} = (G_2(h_0)G_0(h_0) - G_1(h_0)^2)/G_0(h_0)^2$ at a general base point $|h_0| < 1$ (the FDT/linear-response at any perturbation strength), plus the positivity of the perturbed partition there. Result: **a clean fidelity pass, no edits** — with a rejected dead-open flag that, unlike the previous review, was wrong in the *opposite* direction.

1 Setting

This file lifts the $h = 0$ susceptibility (the κ_2 rung) to an arbitrary $|h_0| < 1$, the form the third cumulant needs (the mean’s derivative *as a function of h*). It is the dependency #140’s proof consumed.

2 What was reviewed

Three public theorems: the perturbed-partition positivity, the $n = 0$ partition positivity, and the general- h susceptibility.

3 Fidelity / soundness findings

Sound; GPT found no mismatch.

- `partition_perturbed_pos`: positivity of $\int e^{-t(L+hx)}$ via `integral_pos_iff_support_of_nonneg_ae` — nonnegative a.e. ($e \geq 0$), `support = univ` ($e \neq 0$), $\text{vol}(\text{univ}) = \infty > 0$, and integrability from $|h| < 1$.
- `weightedPartition_zero_pos`: the $n = 0$ specialisation, $G_0(h) = \int e^{-t(L+hx)} > 0$.
- `anharmonic_mean_hasDerivAt_general`: the quotient rule at h_0 , with $f' = G_2$, $g' = G_1$ (general- h partition derivatives) and $G_0(h_0) \neq 0$ from the positivity — giving the expected general- h FDT identity.

4 Simplifications applied

None. GPT’s golf findings (inline `heq/hint/h_support/hc/hd`) were declined. Its dead-open flag — “open `MeasureTheory` unused” — was **rejected**: this file uses the bare integral notation $\int x, \dots$

(e.g. in `partition_perturbed_pos`), which requires `open MeasureTheory`, so the open is *load-bearing*. File left byte-identical.

5 Disagreements and open questions

None on the mathematics.

6 Lessons

- **A reviewer’s dead-open flag is unreliable in *both* directions — two consecutive reviews demonstrate it.** In the third-cumulant review GPT *over*-flagged a load-bearing `open Topology` (it supplies $\mathcal{N}$); here it *under*-recognised a load-bearing `open MeasureTheory` (it supplies the $\int$ integral notation), calling it dead. The discriminator is whether any notation or bare name the open provides is actually used — a per-open grep (and ultimately the build) settles it. “All names are qualified” is not the same as “no notation is used”; integral and neighbourhood notations come from these opens even when every identifier is written long-form.

7 Follow-ups

- The `cumulants-from-log-partition` and the `cumulant-moment-forms` files complete this arc’s review.